

PhiBase: Exact Arithmetic as a Construction

Positions as Counts, and Operations That Run Backward

(working title: *PhiBiz*)

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Abstract

We present an exact integer arithmetic for the golden ratio $\varphi = (1 + \sqrt{5})/2$, built from the object up rather than the algebra down. A position on a line is a record of two step sizes — a unit step of length 1 and a golden step of length φ — and is written as the pair of counts $P(n, p) = n + p\varphi$. Addition, subtraction, and multiplication are closed on these counts (the ring $\mathbb{Z}[\varphi]$); division is exact in the field $\mathbb{Q}(\varphi)$, carried by the homogeneous form $P(n, p, d) = (n + p\varphi)/d$, which keeps its denominator in the open. The organizing property is *reversibility*: each operation runs backward exactly when its operand is retained, and the multiplications that reverse with no denominator and without ever leaving the integer counts are exactly the powers of φ — the Fibonacci spine. That is the whole of the core. Everything the core supports — two lookup tables, a six-count lattice placing dodecahedral geometry in $\mathbb{Z}[\varphi]^3$ with a parity-governed inverse map, a face-plane reflection result (with the correction that these planes are *not* H_3 mirrors), and the applications to function evaluation, parallel lattice computation, and exact-arithmetic model inference — is set out as a study path, with quantitative claims deferred to measurement and to a companion report.

Contents

1 The Construction

Ask a narrow question: what can you reach on a line if you are allowed only two moves — a step of length 1 and a step of length $\varphi = (1 + \sqrt{5})/2$? Every point you can arrive at is a record of how many of each step you took. That record is a pair of integers, and the pair *is* the point. Nothing about this requires a ring to be posited in advance; the algebra is what the counting turns out to obey, not what it is built on.

The single fact that makes the counts arithmetic rather than mere tally is the defining identity of φ ,

$$\varphi^2 = \varphi + 1,$$

which reads, in step language, as: two golden steps equal one golden step plus one unit step. Because of it, any combination of counts — however multiplied — can be rewritten as a plain pair of counts, with no golden step left squared. Section ?? states the representation; Section ?? states the property the whole paper is organized around, that these operations run backward exactly when you keep what you moved by. Everything the core then supports — the tables, the lattice, the geometry, and the applications — is gathered as a study path in Section ??.

The geometry that motivates the choice of φ is dodecahedral and icosahedral: the natural angles are multiples of 36° , and the vertex coordinates of these solids, suitably scaled and oriented, are counts of unit and golden steps [?, ?]. The use of φ as an arithmetic base goes back to Bergman [?]; the contribution here is the constructive reading — positions as counts, operations as reversible moves — and the apparatus that follows from it.

2 Positions Are Counts

Definition 2.1 (Position). A *position* is a pair of integers (n, p) , written $P(n, p)$ and read as the point

$$P(n, p) = n + p\varphi$$

on the line, where n is the number of unit steps and p the number of golden steps. Counts may be negative; a negative count is a step taken backward. The *homogeneous* form $P(n, p, d) = (n + p\varphi)/d$, with d a positive integer, records a position divided by d . The set of positions is $\mathbb{Z}[\varphi] = \{n + p\varphi : n, p \in \mathbb{Z}\}$ [?]; the homogeneous forms range over $\mathbb{Q}(\varphi)$.

Seven positions recur so often as per-axis coordinate values in dodecahedral geometry that they earn single-letter names, listed in Table ??. They are the only values $\{0, \pm 1, \pm \varphi, \pm 1/\varphi\}$ needed to write the vertices of the regular solids built from golden triangles, in the coordinate models used here (Appendix ??).

3 Reversing a Computation

A computation runs forward through the four operations. The question this paper is built around is when it runs *backward*: given a result and the operands, when can an input be recovered exactly, with nothing lost on the way? For counts this is not an existence claim

Table 1: The seven coordinate symbols. (The letter P names the value φ and is unrelated to the position notation $P(n, p)$; context distinguishes them.)

Symbol	$P(n, p)$	Decimal value
z	$P(0, 0)$	0
0	$P(1, 0)$	1
o	$P(-1, 0)$	-1
F	$P(-1, 1)$	$1/\varphi \approx 0.618$
f	$P(1, -1)$	$-1/\varphi \approx -0.618$
P	$P(0, 1)$	$\varphi \approx 1.618$
p	$P(0, -1)$	$-\varphi \approx -1.618$

to wave at — it is a procedure with a cost you can read off. The reverse of addition is subtraction; the reverse of multiplication is division; and each reverse is exact *provided you keep the operand you moved by*. Multiplying a by b and discarding b destroys the information of which pair produced the result, since many pairs share a product. Keep b , and the road back is open: $a = (a \cdot b)/b$ recovers a with no rounding, because you divide out exactly what you multiplied in.

Three observations make this concrete. They are the whole of the machinery.

1. **Closure you can watch.** A count plus a count is a count, and a count times a count is a count. Addition is componentwise,

$$P(n_1, p_1) + P(n_2, p_2) = P(n_1 + n_2, p_1 + p_2),$$

and multiplication, after replacing φ^2 by $\varphi + 1$ (Appendix ??), is

$$P(n_1, p_1) \cdot P(n_2, p_2) = P(n_1 n_2 + p_1 p_2, n_1 p_2 + n_2 p_1 + p_1 p_2).$$

Neither result leaves the integer counts. Subtraction, the reverse of addition, is componentwise and always exact.

2. **The costless reversals are the Fibonacci spine.** Some multiplications undo without keeping any receipt and without ever leaving the counts: exactly those by a position whose reverse is itself a position. These are the positions of norm ± 1 , where $\mathcal{N}(P(n, p)) = n^2 + np - p^2$ (Appendix ??), and they are precisely the signed powers of φ ,

$$\pm \varphi^k = \pm P(F_{k-1}, F_k), \quad k \in \mathbb{Z}, \quad \mathcal{N}(\varphi^k) = (-1)^k,$$

with F_k the k -th Fibonacci number ($F_0 = 0, F_1 = 1, F_{-1} = 1$). This is the Fibonacci spine (Section ??). Multiplying by φ^k is reversed by multiplying by φ^{-k} ; on the spine both are one move along the index, forward by addition and backward by subtraction. The set of free, receipt-less reversals and the central lookup table of the system are the same object.

3. **When the reversal leaves the counts, you keep the divisor.** Every other division carries a denominator, and that denominator is the saved receipt written in the open.

Dividing by b gives

$$\frac{a}{b} = \frac{a \cdot \bar{b}}{\mathcal{N}(b)}, \quad \overline{P(n, p)} = P(n + p, -p),$$

an integer denominator $\mathcal{N}(b)$ under an integer-count numerator — the homogeneous form $P(n, p, d)$ of Definition ???. The denominator d is exactly what you multiply back by to undo the division. Reducing by $\gcd(n, p, d)$ after each step keeps the receipt in lowest terms. Because $\{1, \varphi\}$ is a $\mathbb{Q}$ -basis, two triples denote the same field element only when they are rationally proportional, so the reduced $d > 0$ form is a unique representative; $d = 1$ says the result is back on the integer counts, and $d \neq 1$ says how far off, and by what factor, the return trip still owes.

Remark 3.1 (Why signed counts are needed). The non-negative counts $n, p \geq 0$ are pure forward motion and grow away from the origin. Density near zero comes only from the reverse operator: small positions such as

$$|F_k \varphi - F_{k+1}| = \varphi^{-k}$$

require a step taken backward, e.g. $5\varphi - 8 = \varphi^{-5} \approx 0.09$, which is $P(-8, 5)$. The signed lattice is counting with subtraction admitted, and it is what the Dense Grid (Section ??) tabulates.

3.1 Two Reversals, Worked

A free reversal (on the spine). Take $a = P(1, 1) = 1 + \varphi = \varphi^2$ and multiply by the spine entry $b = P(-1, 1) = \varphi^{-1}$. By the product rule, $a \cdot b = P(0, 1) = \varphi$. To undo it, multiply by $b^{-1} = \varphi = P(0, 1)$: then $\varphi \cdot \varphi = P(1, 1) = a$, recovered exactly, no denominator, never off the counts. The receipt was unnecessary because b was a unit.

A reversal that keeps a receipt. Divide $a = P(1, 0) = 1$ by $b = P(0, 2) = 2\varphi$, with $\mathcal{N}(b) = -4$ and $\bar{b} = P(2, -2)$. The numerator $a \cdot \bar{b} = P(2, -2)$ over -4 reduces to

$$\frac{1}{2\varphi} = P(-1, 1, 2) = \frac{\varphi - 1}{2} \approx 0.309,$$

carrying the receipt $d = 2$. To undo it, multiply back by $b = 2\varphi$: $\frac{\varphi - 1}{2} \cdot 2\varphi = (\varphi - 1)\varphi = \varphi^2 - \varphi = 1 = P(1, 0)$. The denominator was the record of what had been divided out; clearing it is the return trip.

That is the paper's core. The rest is apparatus the core supports.

4 What to Learn Next

The two tables, the lattice, the geometry, and the applications all sit on the core of Section ???. Each is a topic to take up next; below, each is stated far enough to be usable, with its key result and pointers to the fuller algebraic development (the companion manuscript) and the measurement work (the companion report). None of it changes the core.

4.1 The Fibonacci spine

The costless reversals of observation 2, tabulated.

Theorem 4.1 (Powers of φ are counts). *For all $k \in \mathbb{Z}$, $\varphi^k = F_k\varphi + F_{k-1} = P(F_{k-1}, F_k)$.*

Proof. For $k \geq 1$, induction: $\varphi^1 = P(0, 1)$, and if $\varphi^k = P(F_{k-1}, F_k)$ then $\varphi^{k+1} = F_k\varphi^2 + F_{k-1}\varphi = F_k(\varphi + 1) + F_{k-1}\varphi = (F_k + F_{k-1})\varphi + F_k = P(F_k, F_{k+1})$. For $k \leq 0$, use $F_{-n} = (-1)^{n+1}F_n$, starting from $\varphi^{-1} = \varphi - 1 = P(-1, 1)$. $\square$

Each entry is a unit ($\mathcal{N}(\varphi^k) = (-1)^k$), so multiplication and division of exact φ -powers are index addition and subtraction, and one index step forward and one back are inverse moves. The spine extends both ways by a single Fibonacci addition per entry — the lookup-and-add lineage of logarithmic number systems [?, ?], but with algebraically exact entries and no interpolation. The residual scale φ^{-k} (Remark ??) reaches $\approx 10^{-8}$ near $k = 38$, where F_{38} is on the order of 4×10^7 ; that is a statement about the depth of the spine, not the resolution of the bounded grid below.

Table 2: Fibonacci spine entries $\varphi^k = P(F_{k-1}, F_k)$.

k	F_{k-1} (unit count)	F_k (golden count)	φ^k
0	1	0	1.000000
1	0	1	1.618034
2	1	1	2.618034
3	1	2	4.236068
4	2	3	6.854102
5	3	5	11.090170
6	5	8	17.944272
7	8	13	29.034442
8	13	21	46.978714
9	21	34	76.013156

4.2 The dense grid

The spine is sparse and grows away from zero; the dense grid fills the interval with signed counts, whose small values are the near-cancellations of Remark ??. For a bound B , it is the set of positions $P(n, p)$ with $|n|, |p| \leq B$, sorted by value. Two integer identities are kept per entry and must not be confused: the *pair index*, the rectangular address $p N_{\text{cols}} + (n - n_{\text{min}})$, and the *value rank*, the position in sorted order. The counts are exact; only a decimal column is approximate, used solely for nearest-value binary search.

Sizing and resolution follow from B and are read from the generator, not assumed. There are $(2B + 1)^2$ pairs, distinct in value since φ is irrational; at $B = 25$ that is 2601 pairs, of which 1300 are positive. The finest gaps are the residuals φ^{-k} reachable with $|F_k| \leq B$: at $B = 25$ the smallest is $|8\varphi - 13| = \varphi^{-6} \approx 0.013$, so the grid resolves to about 10^{-2} there — not 10^{-8} , which belongs to the spine’s depth. A function is baked in as a map from input

value rank to output value rank, $L_f[r]$; runtime evaluation is one array read after the input is encoded.

4.3 The six-count lattice, and returning by parity

Dodecahedral and icosahedral geometry lifts the counting into three dimensions. A lattice point is six non-negative counts $[a, b, c, d, e, f]$ along the six five-fold-axis directions $\mathbf{B}_1 = (\varphi, 0, 1)$, $\mathbf{B}_2 = (-\varphi, 0, 1)$, $\mathbf{B}_3 = (0, 1, \varphi)$, $\mathbf{B}_4 = (0, -1, \varphi)$, $\mathbf{B}_5 = (1, \varphi, 0)$, $\mathbf{B}_6 = (-1, \varphi, 0)$, giving the forward map into $\mathbb{Z}[\varphi]^3$:

$$x = P(e - f, a - b), \quad y = P(c - d, e + f), \quad z = P(a + b, c + d).$$

This is the projection basis of the Kramer–Neri construction of icosahedral quasicrystals [?]. Writing $x = P(n_x, p_x)$ and so on, the inverse map is $a = \frac{1}{2}(n_z + p_x)$, $b = \frac{1}{2}(n_z - p_x)$, $c = \frac{1}{2}(p_z + n_y)$, $d = \frac{1}{2}(p_z - n_y)$, $e = \frac{1}{2}(p_y + n_x)$, $f = \frac{1}{2}(p_y - n_x)$.

Theorem 4.2 (Return by parity). *A triple $(x, y, z) \in \mathbb{Z}[\varphi]^3$ is the forward image of an integer six-count if and only if $n_z \pm p_x$, $p_z \pm n_y$, and $p_y \pm n_x$ are all even; the six counts are then uniquely recovered by the inverse map.*

The forward map takes counts to space and the inverse takes space back to counts; the $\frac{1}{2}$ in each inverse formula is the receipt, and parity is the test that it clears — the three-dimensional face of the reversal principle. φ -expansion uses $\varphi \cdot P(n, p) = P(p, n + p)$ and returns to an integer six-count on the even-sum sublattice. The full algebraic treatment is in the pinned companion manuscript.

4.4 Reflection in the face planes

Reflect a point through the six face planes of the dodecahedron, whose normals are the five-fold axes. Each normal has $\mathbf{n} \cdot \mathbf{n} = P(2, 1) = \varphi + 2$, with algebraic norm $\mathcal{N}(P(2, 1)) = 5$ and reciprocal $(3 - \varphi)/5$. So reflection maps $\mathbb{Z}[\varphi]^3$ into $\frac{1}{5}\mathbb{Z}[\varphi]^3$; composed with the inverse map’s factor of two, the round-trip denominator is 10, and the reflected point returns to an integer six-count exactly when the numerators (over 5) satisfy $N_z \pm P_x$, $P_z \pm N_y$, $P_y \pm N_x \equiv 0 \pmod{10}$ — once again, a receipt-clearing test.

Remark 4.3 (These are not mirror planes). The six face planes are *not* among the fifteen mirror planes of the icosahedral group H_3 , whose normals lie along the two-fold (edge-midpoint) axes; a reflection through a face plane is not a symmetry of the group. The denominator of 5 each such reflection introduces is the sign of that — a true mirror would permute the vertex set and leave it in $\mathbb{Z}[\varphi]^3$. Constructing the three H_3 generators and extending the reflection tables to all fifteen mirrors, for example to mark hinged edges in a robotic build script, is a separate construction (Section ??).

4.5 The lattice as a data-parallel machine

Because neighbour access and reflection are lookups on precomputed address tables, the geometry maps onto a double-buffered update: each tick dispatches one pass that reads a node’s six neighbours and writes its next state, then swaps buffers.

```
kernel void lattice_step(device Node* last, device Node* next,
                        device int* adj, uint id) {
    next[id] = update(last[adj[id*6+0]], last[adj[id*6+1]],
                      last[adj[id*6+2]], last[adj[id*6+3]],
                      last[adj[id*6+4]], last[adj[id*6+5]]);
}
```

The adjacency and reflection buffers are precomputed once; no coordinate arithmetic runs in the kernel, and the update rule is open (propagation, stress, routing). Products themselves need no multiplier: a $(2N + 1)^2$ table plus adds returns each product, and at model scale the tables are best shared as a codebook keyed by distinct quantized value rather than one per weight. Every such table needs an explicit packed datatype and a proven output range — the product count can exceed a signed byte — and on-chip residency is subject to the device’s threadgroup-memory limit. Throughput is to be measured, not asserted.

4.6 Inference in exact counts

A companion report converts a transformer’s stored weights to positions and runs the matrix arithmetic in exact counts, reproducing reference embeddings on a tested sample [?]. The scope claim to make here is narrow: the weight path is exact by construction, because floating-point code cannot run on tensors stored as integer counts. What the surrounding operations (normalization, attention, activation) run in is a separate question the companion report addresses directly.

4.7 Energy, as an open and measurable question

Exact reversible arithmetic retains rather than erases intermediate information. In the idealized Landauer picture [?, ?], information-preserving operations carry no thermodynamic lower bound on dissipation, unlike the bit erasure implicit in rounding. Whether that yields measurable savings on real hardware is empirical: the honest comparison is per-operation energy on the target device against a floating-point baseline [?], and it must be measured. No fixed multiplier is claimed on paper.

4.8 Further geometry and layout

Two open directions. First, the fifteen H_3 mirrors: constructing the three generators (normals perpendicular to the two-fold axes) and extending the reflection tables to all fifteen planes supports hinged-edge build operations, and is separate from the face planes of Section ???. Second, golden-angle layout: the counts that place polyhedral vertices also place phyllotactic points, and whether the reversal bookkeeping simplifies golden-angle layouts for arrays and sensor placement is open.

A Arithmetic on the Counts

With $\varphi^2 = \varphi + 1$,

$$(n_1 + p_1\varphi)(n_2 + p_2\varphi) = (n_1n_2 + p_1p_2) + (n_1p_2 + n_2p_1 + p_1p_2)\varphi.$$

The conjugate sends $\varphi \mapsto 1 - \varphi$, so $\overline{P(n, p)} = P(n + p, -p)$, and the norm

$$\mathcal{N}(P(n, p)) = P(n, p) \cdot \overline{P(n, p)} = n^2 + np - p^2 \in \mathbb{Z}$$

is multiplicative. Division uses the conjugate: $a/b = a\bar{b}/\mathcal{N}(b)$. A position is a unit ($1/P(n, p) \in \mathbb{Z}[\varphi]$) iff $\mathcal{N}(P(n, p)) = \pm 1$, and the units are exactly $\pm\varphi^k$.

B The Coordinate Vocabulary

The seven symbols of Table ?? span $\{0, \pm 1, \pm\varphi, \pm 1/\varphi\}$. In the coordinate models used here, the dodecahedral and icosahedral vertex sets can be scaled and oriented so that every coordinate component is one of these seven values; a 3D vertex is then a three-symbol string (e.g. $000 = (1, 1, 1)$, $\mathbf{zfP} = (0, -1/\varphi, \varphi)$).

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